

Exploratory Data Analysis (EDA) Report

Geldium Delinquency Risk Assessment Project

Prepared for:

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Objective

The purpose of this exploratory data analysis (EDA) is to assess the quality, completeness, and reliability of Geldium's customer financial dataset before predictive modeling. The analysis focuses on identifying missing or inconsistent data, detecting potential risk indicators related to delinquency, and recommending appropriate data treatment strategies to support future AI-driven risk assessment models.

Step 1: Dataset Review and Initial Insights

Key Findings

Missing or Inconsistent Data

- Missing values were identified in important financial variables such as income, payment history, and credit utilization.
- Some customer records contained incomplete repayment information, which may affect delinquency prediction accuracy.
- A small number of duplicate customer records and inconsistent formatting issues were observed.
- Certain variables showed unusual values, including extremely high credit utilization rates and irregular payment entries.

Key Anomalies

- Some customers with high credit scores also showed repeated missed payments, indicating possible inconsistencies or hidden financial stress.
- Outlier values were observed in debt balances and credit utilization percentages.
- A few customer profiles displayed unusually rapid increases in credit activity, which may signal elevated financial risk.

Early Indicators of Delinquency Risk

- Customers with repeated late payments demonstrated a significantly higher likelihood of delinquency.
- High credit utilization appeared strongly associated with repayment difficulties.
- Customers with unstable income or multiple recent credit applications showed elevated risk patterns.
- High debt-to-income ratios indicated potential financial stress and repayment challenges.

Initial Data Quality Summary

The dataset provides several valuable financial and behavioral indicators relevant to delinquency prediction; however, some data quality concerns were identified. Missing values in key financial fields, duplicate records, and inconsistent formatting may reduce prediction reliability if not addressed properly. High credit utilization, irregular repayment patterns, and elevated debt levels emerged as important early indicators of delinquency risk. Overall, the dataset is suitable for predictive modeling after appropriate data cleaning, imputation, and validation processes are completed.

Step 2: Missing Data and Data Quality Treatment

Missing Data Issue	Handling Method	Justification
Missing income values	Median imputation	Median values reduce the effect of outliers and maintain realistic income distributions.
Missing credit utilization records	Regression/statistical imputation	Credit utilization is an important risk factor and should be estimated using related financial variables instead of deletion.
Incomplete payment history	Synthetic data generation with validation	Synthetic records can supplement missing behavioral data while preserving privacy and maintaining historical distribution patterns.

Additional Data Quality Actions

- Duplicate customer records were identified and removed.
- Inconsistent formatting in financial fields and date columns was standardized.
- Extreme outlier values were reviewed to determine whether they reflected legitimate high-risk behavior or data entry errors.

Step 3: Risk Factors and Pattern Detection

High-Risk Indicators

Repeated Late Payments

Customers with a history of missed or delayed payments showed the highest likelihood of future delinquency.

High Credit Utilization

Customers using a large percentage of their available credit demonstrated increased financial stress and elevated repayment risk.

High Debt-to-Income Ratio

A high DTI ratio suggests that customers may struggle to manage existing financial obligations.

Sudden Increase in Credit Activity

Multiple recent loan applications or new credit accounts may indicate financial instability.

Income and Employment Instability

Customers with inconsistent income patterns or unstable employment histories may face greater repayment difficulties.

Additional Insights

- Customers with one missed payment frequently demonstrated repeated delinquency behavior.
- Younger customers with limited credit histories appeared to have slightly higher delinquency tendencies.
- Customers with increasing outstanding balances over time may require proactive intervention strategies.
- Some anomalies identified in high-credit-score customer segments may require further investigation by the analytics team.

Conclusion

The exploratory data analysis identified several important data quality issues and financial risk indicators relevant to delinquency prediction. Missing values, duplicate records, and inconsistent financial information require preprocessing before predictive modeling can be implemented. Key delinquency indicators include repeated late payments, high credit utilization, unstable income, and elevated debt-to-income ratios. Appropriate imputation strategies and carefully validated synthetic data generation methods can improve dataset completeness while maintaining fairness, privacy, and regulatory compliance.

The findings from this analysis will support Tata iQ and Geldium in developing a more reliable, explainable, and AI-driven delinquency prediction framework for proactive customer intervention and improved risk management.